

Information Retention Across Geometry-Attached Observations in the Tri-Octagon Kernel

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Abstract

We determine what existing geometry-attached observations retain from a native complex triad. The complete labelled face-vector decoder is an inherited invertible representation, whereas lengths, populations, signed areas and the axial vector select different information. On the published positive SPR-01 one-update family, all signed areas and the axial vector vanish exactly, yet raw intensity is globally injective in the amplitude. A published population collision retains opposite signed coherence in the existing symmetric pair matrix. The complete mathematical axial snapshot is richer than Hermitian coherence because its quadrature accounting supplies the complex bilinear quantity $w^T w$. We prove its static fibres: an overall-sign pair when this quantity is nonzero, an entire common-phase circle for nonzero isotropic states, and a singleton at zero. Previously executed comparisons on 205 inputs and nine static controls corroborate the unchanged implementation within a fixed numerical criterion. Exact mathematics, finite execution evidence and unresolved measurement questions remain separate. Geometry creates no new input information, and no generic inverse, full-record dynamical closure, physical realization or computational advantage is established.

1 Question, foundations and scope

When a display cannot distinguish two native states, is the distinction absent from the state, lost by projection, discarded by normalization, or retained elsewhere in the existing observation record? This paper answers that question for declared observations, using one accepted input family and fixed static controls. “Retains” means exact mathematical identifiability unless explicitly qualified as a finite numerical comparison.

Native Response Geometry (NRG), incorporating the completed SA0 attachment analysis, supplies the decoder, representation/observation distinction and real-Gram fibre result [1]. Paper G supplies the signed pair algebra and rank-two axial attachment [2]. Published SPR-01 supplies the one-update family, population collision and raw/coherence inverses [3]. SGO-01 applies these inherited results, adds the intensity and complete-record deductions below, and supplies the retained bounded comparisons [7]. None is a literature-wide novelty claim.

The labels *inherited result*, *SGO-01 exact specialization/deduction*, *bounded numerical corroboration* and *open operational/physical question* distinguish the sources of claims. Review and reciprocal checking were project-internal, not external peer review. This edition implies no journal acceptance, DOI, software release or physical validation.

2 Adopted face representation

Inherited result. Write $w = x + iy \in \mathbb{C}^3$, with real channel triples x, y ordered A,B,C. In the fixed canonical shell, outward unit normals and horizontal tangents are

$$\begin{aligned} n_A &= (-\sqrt{3}/2, 1/2, 0), & t_A &= (-1/2, -\sqrt{3}/2, 0), \\ n_B &= (0, -1, 0), & t_B &= (1, 0, 0), & t_i &= e_z \times n_i. \\ n_C &= (\sqrt{3}/2, 1/2, 0), & t_C &= (-1/2, \sqrt{3}/2, 0), \end{aligned} \quad (1)$$

The centres are $c_A = (-1/4, \sqrt{3}/4, 0)$, $c_B = (0, 0, 0)$ and $c_C = (1/4, \sqrt{3}/4, 0)$. The complete labelled representation and its encoder are

$$v_i = D_i(w_i) = x_i t_i + y_i e_z, \quad E_i(v) = t_i \cdot v + i e_z \cdot v. \quad (2)$$

Orthonormality of (t_i, e_z) gives

$$E_i D_i = \text{id}, \quad D_i E_i = I_3 - n_i n_i^T, \quad q_i := \|v_i\|^2 = |w_i|^2. \quad (3)$$

Thus ED is identity on $\mathbb{C}^3$, whereas DE is only a tangent projector on ambient arrays. With labels and frames retained, the three face vectors retain all six real state coordinates, including scale and common phase [1, 5].

SGO-01 / canonical shell, geometry only

Actual folded_module() mesh · transparent faces · shared seams emphasized · no state arrows

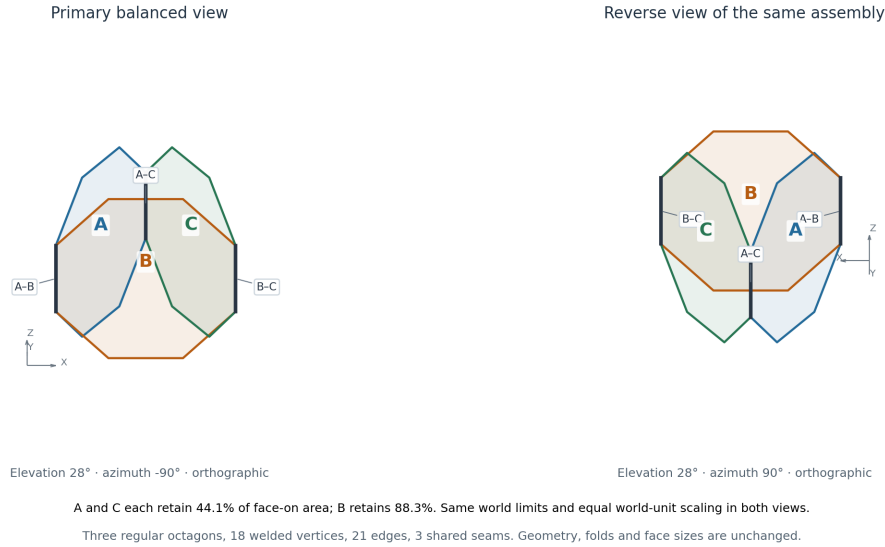


Figure 1: Accepted v02 geometry-only views, using the actual `geometry.folded_module()` vertices and face cycles. A, B and C are congruent canonical regular octagons with welded, labelled shared seams. The primary and reverse cameras view the same unchanged assembly. Transparency exposes rear seams; projected face areas are not physical face sizes. There are no state arrows or deformations in this figure.

The vectors in (2) are free vectors based at fixed centres, not vertex motion, particle position, a field or a second recurrence. The internal `FaceState.step` advances stored canonical w and then decodes it. It does not repeatedly evolve a rounded $E(D(w))$. Its stored state is used only for execution parity; vector reconstruction uses vectors plus frames alone.

3 Matched pair data and observation hierarchy

Inherited result. Define symmetric and antisymmetric pair matrices, Hermitian coherence and channel area by

$$S = xx^T + yy^T, \quad A = xy^T - yx^T, \quad H = ww^\dagger = S - iA, \quad (4)$$

$$C = x \times y = (A_{BC}, A_{CA}, A_{AB})^T. \quad (5)$$

The sign in $H = S - iA$ distinguishes it from the conjugate coherence. Channel-area C is not an ambient face vector.

Inherited result. The matched source- j to destination- i transport is

$$T_{ij} = t_i t_j^T + e_z e_z^T, \quad T_{ij} v_j = x_j t_i + y_j e_z. \quad (6)$$

Consequently the geometric pair contractions are

$$v_i \cdot (T_{ij} v_j) = S_{ij}, \quad n_i \cdot (v_i \times T_{ij} v_j) = A_{ij}. \quad (7)$$

The second identity is the inherited signed parallelogram-area convention: $t_i \times e_z = n_i$, with no factor $1/2$ [2]. The explicit matched-dot readout is an SGO-01 specialization of the adopted transport; it is an externally calculated mathematical contraction, not a new named API or a detector.

Raw ambient dot products differ: $v_i \cdot v_j = (t_i \cdot t_j)x_i x_j + y_i y_j$. For distinct faces $t_i \cdot t_j = -1/2$. At $(21, 2, 21)$, for example, the untransported AB dot is -21 , whereas $S_{AB} = 42$.

Observation	Generic static retention/loss	Identifies positive family amplitude?
Labelled $D(w)$	Entire state in the adopted frames	Yes
$q; I = \sum q_i$	Magnitudes and scale; norm only	Yes; yes
q/I	Normalized magnitudes, no scale or phases	No
A or C	Oriented area, not full state or intensity	No: zero
W	Planar projection of C	No: zero
(W, Γ)	All of C , not the state	No: zero
S	Real pair coherence, up to phase/conjugation	Yes
(S, A) or H	Nonzero state modulo common phase	Yes
$S/I; H/I$	Also lose scale	Yes; yes
Complete snapshot	Equivalent to $(H, w^T w)$; see Section 8	Yes

Table 1: Exact information contracts. The family column refers only to the fixed $a > 0$ family in Section 5; it is not generic inversion. Named normalized bundles require $I > 0$. Numerical residuals and metadata are excluded as information channels.

4 Generic static losses and the axial projection

4.1 Magnitudes, real Gram and complex coherence

Inherited result. Magnitudes $q_i = |w_i|^2$ are invariant under independent channel phases; total intensity retains only the norm. Division by I also loses positive scale. These statements do not imply that the same losses persist on every restricted family.

The generic real-Gram fibre characterization is inherited from NRG's regular real-Gram closure proof, including its rank-deficient factor argument [1]. For completeness, write $B = [x \ y]$, so $S = BB^T$. Equal row Gram matrices determine an isometry of the spanned row spaces, extending to an $O(2)$ transformation: $B' = BO$. Rotations give common phase, and reflections give common phase followed by conjugation. Hence

$$S(w') = S(w) \iff w' = e^{i\theta}w \text{ or } w' = e^{i\theta}\bar{w}. \quad (8)$$

This inherited result is applied here, not claimed as a new SGO-01 theorem. For nonzero $H = ww^\dagger$, equality fixes the same rank-one complex range, so $w' = cw$ with $|c| = 1$. Thus H retains the state modulo common phase; H/I retains only its complex line, with arbitrary $c \neq 0$. Normalized S/I adds positive-scale ambiguity to (8). Zero has a singleton unnormalized S or H fibre.

These are static information statements. NRG's regular closure results require nonzero pre-synchronization components; the native zero convention supplies a separate obstruction. A static fibre description alone is not a full-domain autonomous update law.

4.2 What the axial vector discards

Inherited result. Let $e = (1, 1, 1)^T$. Paper G's ambient attachment uses

$$T = \begin{pmatrix} -1/2 & 1 & -1/2 \\ -\sqrt{3}/2 & 0 & \sqrt{3}/2 \\ 0 & 0 & 0 \end{pmatrix}, \quad W = TC, \quad \Gamma = e^T C. \quad (9)$$

This T , with columns t_i , differs from T_{ij} . Its column inner products give

$$T^T T = \frac{3}{2} \left(I_3 - \frac{ee^T}{3} \right), \quad \ker T = \mathbb{R}e, \quad C = \frac{2}{3} T^T W + \frac{\Gamma}{3} e. \quad (10)$$

Thus (W, Γ) recovers C , whereas $W = 0$ only implies $C \parallel e$. Neither observation generally determines the state or its intensity. For example, $x \mapsto tx$, $y \mapsto y/t$ preserves C for nonzero real t while generally changing intensity. Also $C = 0$ exactly when x, y are collinear, including every real state.

SGO-01 exact specialization/deduction. The prescribed static control $v = (1, -1 + i, -i)$ has $x = (1, -1, 0)$, $y = (0, 1, -1)$, hence $C = (1, 1, 1)$, $W = 0$ and $\Gamma = 3$. It exhibits area hidden by the projection, not an evolved trajectory or a physical field effect. This applies the inherited reconstruction theorem rather than replacing it.

5 The fixed SPR-01 one-update family

Inherited result. The native triad update first forms an amplitude prestage and then synchronizes phases simultaneously:

$$u = z + \epsilon z \odot (k - |z|^2) + gLz, \quad L = ee^T - 3I_3, \quad (11)$$

$$F_i(z) = |u_i|e^{i(\phi_i + \delta_i)}, \quad \delta_i = \lambda \sum_{j \neq i} \sin(3(\phi_j - \phi_i)), \quad (12)$$

where $\phi_i = \text{Arg}_0(u_i)$, $\text{Arg}_0(0) = 0$, and zero prestage components produce zero outputs [4, 5]. There is no normalization or geometric feedback in this recurrence.

The accepted SPR-01 settings are

$$z(a) = a(1, 2, 1)^T, \quad a > 0, \quad \epsilon = \frac{1}{20}, \quad g = \frac{1}{5}, \quad \lambda = \frac{1}{30}, \quad k = (1, 1, 1). \quad (13)$$

One update is applied to each independently prepared input. Substitution into (11) gives the inherited exact response

$$w(a) = F(z(a)) = \frac{a}{20}(25 - a^2, 34 - 8a^2, 25 - a^2)^T. \quad (14)$$

All nonzero prestage phases are 0 or π , so the threefold sine increments vanish exactly. The zero convention leaves any cancelled component zero. Thus (14) is the full update, not only the prestage [3].

SGO-01 exact specialization/deduction. Set $X = a^2$, $s = 25 - X$, $r = 17 - 4X$ and $d = s^2 + 2r^2$. The existing face and pair observations specialize to

$$D(w) = \begin{pmatrix} -as/40 & -\sqrt{3}as/40 & 0 \\ ar/10 & 0 & 0 \\ -as/40 & \sqrt{3}as/40 & 0 \end{pmatrix}, \quad q = \frac{a^2}{400}(s^2, 4r^2, s^2)^T, \quad (15)$$

$$S = H = \frac{a^2}{400} \begin{pmatrix} s^2 & 2sr & s^2 \\ 2sr & 4r^2 & 2sr \\ s^2 & 2sr & s^2 \end{pmatrix}, \quad A = C = W = \Gamma = 0. \quad (16)$$

The two factors s, r cannot vanish together, so $I = a^2d/200 > 0$ for every $a > 0$. Dividing (15) and (16) by I gives $q/I = (s^2, 4r^2, s^2)^T/(2d)$ and the displayed matrix divided by $2d$. Also $A/I = 0$. A zero axial arrow is therefore entirely uninformative about amplitude on this family, while the complete vectors remain nonzero. Quadrature accounting has $Q_x = I$, $Q_y = h = 0$; its meaning is developed in Section 8.

6 Intensity injectivity and inverse domains

Theorem 1 (SGO-01 exact deduction: positive-family intensity). *The inherited output-intensity polynomial*

$$I(a) = \frac{a^2(33a^4 - 322a^2 + 1203)}{200} \quad (17)$$

is a bijection from $a > 0$ to $I > 0$.

Proof. Differentiation and completing the square give

$$I'(a) = \frac{a}{100} \left[99 \left(a^2 - \frac{322}{99} \right)^2 + \frac{15413}{99} \right] > 0. \quad (18)$$

Also $I(a) \rightarrow 0$ as $a \rightarrow 0^+$ and $I(a) \rightarrow \infty$ as $a \rightarrow \infty$. Continuity and strict monotonicity prove the claim. $\square$

Given $I > 0$, there is a unique positive solution X of

$$33X^3 - 322X^2 + 1203X = 200I, \quad a = \sqrt{X}. \quad (19)$$

Consequently raw q , S and the full snapshot also identify a . This is exact identifiability on a known family, not generic sign detection, an inverse-conditioning estimate or a finite-noise theorem. Generic states $(21, 2, 21)$ and $(21, -2, 21)$ have equal intensity but different relative sign. Entrance intensity $6a^2$ already identifies positive a ; geometry does not create that information.

Inherited result. The SPR-01 raw inverse and normalized-coherence chart are

$$a = \frac{10}{83}(8w_A - w_B), \quad t = \frac{(S/I)_{BA}}{2(S/I)_{AA}} = \frac{17 - 4X}{25 - X}, \quad a = \sqrt{\frac{17 - 25t}{4 - t}}. \quad (20)$$

The chart requires $a \neq 5$. Its projective injectivity follows from

$$(25 - X)(17 - 4Y) - (17 - 4X)(25 - Y) = 83(X - Y). \quad (21)$$

At $a = 5$, the unique pure-B normalized direction identifies the exceptional point. Since $S = H$ on this real family, S/I alone suffices [3].

Input	Exact output and intensity	Normalized status
$a = 0$	$w = 0, I = 0$	Undefined; outside $a > 0$
$a = \sqrt{17}/2$	$w = (83\sqrt{17}/160, 0, 83\sqrt{17}/160);$ $I = 117113/12800$	$q/I = (1/2, 0, 1/2);$ $S/I = \frac{1}{2} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{pmatrix}$
$a = 5$	$w = (0, -83/2, 0); I = 6889/4$	$q/I = (0, 1, 0); S/I = \text{diag}(0, 1, 0)$

Table 2: Zero and component cancellations. $A/I = 0$ at both nonzero cancellations. These exact family values do not extend the earlier general nonzero-prestage closure theorem through its excluded points.

7 Same populations, different retained information

Inherited result. SPR-01's collision uses two unequal-scale outputs, at

$$a = 2, \quad b = \sqrt{382/85}, \quad \kappa = \frac{83b}{1700}, \quad w(2) = \frac{1}{10}(21, 2, 21), \quad w(b) = \kappa(21, -2, 21). \quad (22)$$

Let

$$M_{\pm} = \begin{pmatrix} 441 & \pm 42 & 441 \\ \pm 42 & 4 & \pm 42 \\ 441 & \pm 42 & 441 \end{pmatrix}. \quad (23)$$

SGO-01 exact specialization/deduction. The complete raw/readout comparison specializes the inherited collision as follows:

Readout	$a = 2$	$a = b$
q	$(441, 4, 441)/100$	$\kappa^2(441, 4, 441)$
I	$443/50 = 8.86$	$582898957/61412500$
$S = H$	$M_+/100$	$\kappa^2 M_-$
q/I	$(441, 4, 441)/886$	$(441, 4, 441)/886$
$S/I = H/I$	$M_+/886$	$M_-/886$
$(S/I)_{AB}$	$+21/443$	$-21/443$
A, C, W, Γ	All zero	All zero
(Q_x, Q_y, h)	$(443/50, 0, 0)$	$(582898957/61412500, 0, 0)$

Table 3: Actual ideal outputs of the two published amplitudes, not the equal-scale sign controls. The vectors are $D(w(2))$ and $D(w(b))$ under the same fixed decoder.

The second intensity is approximately 9.49153603908, and

$$I(b) - I(2) = \frac{38784207}{61412500} > 0. \quad (24)$$

Thus identical normalized populations do not imply equal raw lengths or intensity. The opposite signed coherence survives in the already-existing S field, in H , in their normalized versions, in the full snapshot and in matched geometric dot products. It is not stored in A , which vanishes for both ideal real outputs.

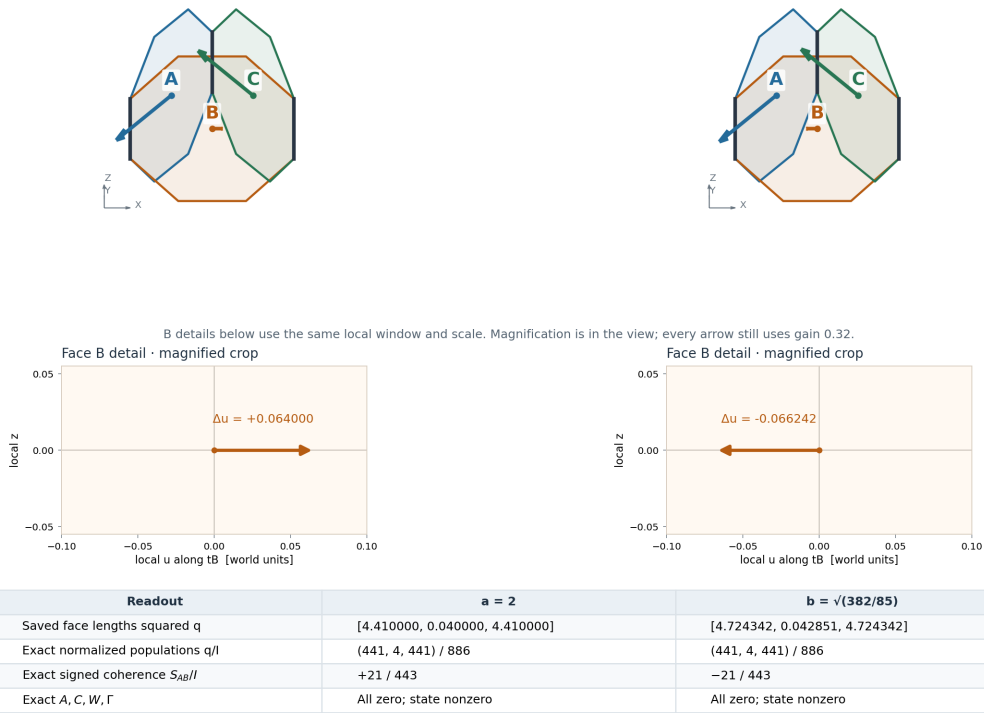
The fixed equal-scale controls $r_+ = (21, 2, 21)$ and $r_- = (21, -2, 21)$ isolate a different point: both have $q = (441, 4, 441)$ and $I = 886$, but $S = M_+$ versus M_- . This demonstrates loss of relative sign by magnitude-only observation without attributing generic sign detection to raw intensity. Figure 2 uses the actual unequal-scale native outputs, preserving their saved floating-point components.

SGO-01 / saved outputs on the canonical shell

Same geometry, camera, world scale and arrow gain 0.32 · no new native update

$$a = 2 \quad l = 8.860000000$$

$$b = \sqrt{382/85} \quad l = 9.491536039$$



Full panels: orthographic camera (28°, -90°), identical world limits; dark segments are shared seams. Face vectors are free vectors at fixed centres.
 Both normalized outputs are defined. At $a = 0$ normalization is undefined (not shown). Tiny binary64 residues remain in the saved data and arrows.

Figure 2: Accepted v02 comparison of two separately prepared one-update outputs, not successive frames. The top panels use the same canonical regular-octagon assembly, welded shared seams, orthographic camera, world scaling and arrow gain 0.32. Arrows are free state vectors at fixed face centres, not deformation, force, field lines or particle displacement. The labelled B-detail panels are magnified local crops with identical limits and unchanged arrow gain, making the sign reversal readable. Saved binary64 residues are retained; exact zero-area labels describe the ideal real family. The full-resolution PNG accompanies this edition.

8 Complete snapshot information and phase qualification

Inherited result. The existing public `AxialSnapshot` includes S, A, C, W, Γ, I , area decomposition and norms, quadrature accounting, residuals and constant metadata [5]. It is not merely W , and it is not a squared-magnitude-only record. In particular,

$$Q_x = \|x\|^2, \quad Q_y = \|y\|^2, \quad h = x \cdot y, \quad I = Q_x + Q_y. \quad (25)$$

These are the accounting fields named **A**, **B** and **h**; the first is not the antisymmetric matrix A .

Field group	Exact information content
S, A	$H = S - \mathbf{i}A$
$C, \Gamma, C_{\parallel}, C_{\perp}, W$	C from A ; $\Gamma = e^T C$; $C_{\parallel} = \Gamma e/3$; $C_{\perp} = C - C_{\parallel}$; $W = TC$
Intensity and area/axial norms	$I = \text{tr } S$, $\ C\ $, $\ W\ $
Accounting Q_x, Q_y, h	Additional quadrature-reference information
Accounting Gram and slack fields	$Q_x Q_y, h^2, Q_x Q_y - h^2 = \ C\ ^2$; $I/2$; $(Q_x - Q_y)^2/4 + h^2 = I^2/4 - \ C\ ^2$
Duplicate accounting fields	$C, I, \ C\ , \ C\ ^2$
Residuals	Zero in exact mathematics; finite-arithmetic diagnostics
Revision/source/policy metadata	Constants, not state information

Table 4: The entire mathematical record reduces to H and (Q_x, Q_y, h) . Heterogeneous metadata and numerical residuals are not normalized or treated as hidden recovery channels.

The residuals express $A + [C]_{\times} = 0$, $SC = 0$, (10), $W_z = 0$, and

$$e_2(S) = \|C\|^2, \quad I^2 - |w^T w|^2 = 4\|C\|^2, \quad \|C\|^2 = \frac{2\|W\|^2 + \Gamma^2}{3}, \quad (26)$$

as well as the Gram/slack equalities in Table 4. Here $[C]_{\times} v = C \times v$ and e_2 sums principal 2×2 minors.

SGO-01 exact specialization/deduction. The older shorthand “snapshot common-phase invariance” must be restricted to the selected invariant fields S, A, C, W, Γ, I [6]. It does not describe the entire heterogeneous record: at r_+ the accounting pair is $(Q_x, Q_y) = (886, 0)$, whereas at ir_+ it is $(0, 886)$, despite identical H . This is a documentation-scope qualification; no runtime field is removed or changed. NRG’s regular H/S closure result cannot automatically be promoted to full-record closure.

8.1 Static fibres of the complete mathematical record

Theorem 2 (SGO-01 exact deduction: complete-record fibres). *Excluding constant metadata and floating residuals, the snapshot retains exactly the information in (H, β) , where*

$$\beta(w) = w^T w = Q_x - Q_y + 2ih. \quad (27)$$

Its fibre at w is $\{w, -w\}$ if $\beta(w) \neq 0$; it is $\{e^{i\theta}w : \theta \in \mathbb{R}\}$ if $w \neq 0$ and $\beta(w) = 0$; and it is $\{0\}$ at zero.

Proof. Table 4 shows that the whole record is a function of H, Q_x, Q_y, h , which conversely occur among its fields. Since H supplies $I = Q_x + Q_y$, (27) recovers $Q_x = (I + \Re\beta)/2$, $Q_y = (I - \Re\beta)/2$, $h = \Im\beta/2$.

For $w \neq 0$, any state with equal H has form $w' = e^{i\theta}w$. But $\beta(w') = e^{2i\theta}\beta(w)$. If $\beta \neq 0$, equality requires $e^{2i\theta} = 1$, giving exactly w and $-w$. If $\beta = 0$, then $Q_x = Q_y = I/2$ and $h = 0$, unchanged by every common phase. Finally $H = 0$ forces $w = 0$. $\square$

The additional phase reference is a reference in the supplied complex coordinates, not a physical apparatus. These fibres are static; the theorem is neither an inverse of F nor a complete-snapshot dynamical closure law.

8.2 The nine fixed static controls

SGO-01 exact specialization/deduction. Let $q_r = (441, 4, 441)$ and use $M_{\pm}$ from (23). All prescribed controls in Table 5 are unevolved observer inputs. Their exact normalized bundles follow by dividing q, S, A individually by the nonzero I .

State	$q; I$	$S; A$	$C; \Gamma$	(Q_x, Q_y, h)
r_+	$q_r; 886$	$M_+; 0$	$0; 0$	$(886, 0, 0)$
r_-	$q_r; 886$	$M_-; 0$	$0; 0$	$(886, 0, 0)$
$-r_+$	$q_r; 886$	$M_+; 0$	$0; 0$	$(886, 0, 0)$
ir_+	$q_r; 886$	$M_+; 0$	$0; 0$	$(0, 886, 0)$
$2r_+$	$4q_r; 3544$	$4M_+; 0$	$0; 0$	$(3544, 0, 0)$
$u = (1, i, 0)$	$(1, 1, 0); 2$	$S_u; A_u$	$(0, 0, 1); 1$	$(1, 1, 0)$
$\bar{u}$	$(1, 1, 0); 2$	$S_u; -A_u$	$(0, 0, -1); -1$	$(1, 1, 0)$
iu	$(1, 1, 0); 2$	$S_u; A_u$	$(0, 0, 1); 1$	$(1, 1, 0)$
$v = (1, -1 + i, -i)$	$(1, 2, 1); 4$	$S_v; A_v$	$(1, 1, 1); 3$	$(2, 2, -1)$

Table 5: Exact static controls. $W = TC$ throughout. The complex controls isolate conjugation, an isotropic common-phase ambiguity and area hidden by the planar projection.

8.3 Explicit control matrices and interpretation

The matrices used in Table 5 are

$$S_u = \text{diag}(1, 1, 0), \quad A_u = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad W(u) = (-1/2, \sqrt{3}/2, 0), \quad (28)$$

$$S_v = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}, \quad A_v = \begin{pmatrix} 0 & 1 & -1 \\ -1 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix}. \quad (29)$$

The controls separate loss of relative sign (r_+, r_-) , global sign $(r_+, -r_+)$, quadrature-reference phase (r_+, ir_+) , scale under normalization $(r_+, 2r_+)$, and area orientation $(u, \bar{u})$. Since $u^T u = 0$, u and iu have the same entire mathematical snapshot, even though their complete face vectors differ. For v , nonzero C is hidden in W but recovered by Γ . These are applications of the explicit formulas and Theorem 2; no additional evolved controls were introduced.

9 Bounded implementation evidence

Bounded numerical corroboration. The completed study froze the following comparison before execution [7]:

$$a_j = \frac{1}{2} + \frac{j}{100}, \quad j = 0, \dots, 200, \quad \text{plus } \sqrt{17}/2, \sqrt{382/85}, \quad (30)$$

giving 203 main-window inputs. The separately flagged diagnostics $a = 0, 5$ give 205 distinct inputs. Each was evaluated once through each of three existing paths: native `step3` followed by face/axial observation; internal `FaceState.step(config, profile=None)`; and public `api.step(State(...), Parameters(...), topology="triad")` followed by `api.axial_snapshot`. This yields 615 update-route evaluations, not 615 distinct inputs. Nine static controls through native/public observers add 18 records without updates.

An independent exact/100-digit reference used explicit radical frames rather than the rounded frames being tested. Three error levels were separated: state error against ideal $w(a)$; observer error against formulas on the exact supplied binary64 components; and end-to-end readout error against ideal $w(a)$. Binary64 components were lifted by integer ratios. Each named field used the frozen criterion

$$\text{error} \leq 10^{-12} + 10^{-10} \|\text{reference}\|, \quad (31)$$

with absolute scalar, Euclidean vector and Frobenius matrix norms. Undefined normalized fields at zero were null, not zero. No phase alignment, clipping, forced zeros, sign repair or tolerance retuning occurred. This criterion is numerical, not a physical measurement-noise assumption.

All 205 route triples had byte-identical canonical states and equal named observations; all nine static native/public pairs agreed. The 204 inputs shared with original SPR-01 reproduced its saved native/public output bytes. No checked-arithmetic refusal occurred. These are finite execution findings, not uniform guarantees outside the fixtures or installed-UI certification.

9.1 Recorded errors, independent audit and passivity

Bounded numerical corroboration. Across the 203 main-window inputs, the maximum raw native-state error was $1.7281369134961648 \times 10^{-15}$. Representative field maxima are shown in Table 6; the other routes agree. The 633 records each contain a state and 48 named readouts/diagnostics.

Field	Observer-only error	End-to-end error
Complete face vectors	4.5221×10^{-16}	1.7257×10^{-15}
q vector	2.4522×10^{-15}	6.5431×10^{-15}
Snapshot intensity	1.5550×10^{-15}	9.2304×10^{-15}
S	8.8730×10^{-16}	1.0751×10^{-14}
A	1.3555×10^{-31}	1.2629×10^{-15}
C	9.5847×10^{-32}	8.9302×10^{-16}
W	1.5982×10^{-31}	1.0937×10^{-15}
Matched dot matrix	4.3196×10^{-15}	1.0364×10^{-14}
q/I	1.2983×10^{-16}	4.3157×10^{-16}
S/I	1.6697×10^{-16}	6.0066×10^{-16}

Table 6: Previously executed main-window maxima under the fieldwise norms in (31). Observer-only and propagated state errors are not conflated. Small numerical area values do not contradict exact ideal-family zeros.

Their 61,401 comparisons comprise 61,383 finite inequalities and 18 structurally undefined/null checks; all pass. There are also 62 supporting exact predicates. These heterogeneous counts are not theorem counts, independent trials or hardware certificates.

The largest recorded error/limit fraction was 0.2109147949831529, for end-to-end accounting h at $a = 5$. There the stored output is approximately $(0, -41.5 + 5.082284216461516 \times 10^{-15}i, 0)$, giving $h = -2.109147949831529 \times 10^{-13}$ rather than its ideal zero. At b , A_{AB} is about 6.0612×10^{-17} and W_y about 1.0498×10^{-16} , again versus exact zeros. These residues were retained, not interpreted as new information.

The later project-internal saved-data audit independently reconstructed the static formulas at 110-digit precision, without importing the original verifier helpers or kernel modules [8]. It reconciled all field metrics, maxima and case locators, all 31,017 CSV rows, route equality, controls and the SPR-01 overlap. This is independent checking of saved data, not a replay of the native or public implementation.

The original execution recorded passive observations preserving supplied state/vector bytes and public indices, with no evolution during observation. Recorded calls were 615 native triad updates, 615 phase synchronizations, zero ring updates, 633 explicit encoder checks, and 419 native-owner/214 public-facade snapshots. No encoder feedback occurred within FaceState evolution. These historical profiler and preservation records remain execution evidence, not events independently reenacted by the later saved-data audit. Publication preparation performs no new scientific execution.

10 Comparators, limits and conclusions

Inherited result. The negative RFO-02 classification result remains unchanged: under its own fixed setup and readout-error assumptions, the chosen inputs were already separable at the entrance and the nonlinear path lost the distinction by step 12 [3, Section 9]. The present one-update family and more informative existing observations do not overturn that experiment.

The proper comparators are the same raw canonical state, directly calculated I, S, A , the original population-only readout, and entrance intensity $6a^2$. If the amplitude is already supplied numerically, direct evaluation of the formulas yields the same mathematical readouts. Geometry changes representation and which coordinates a display selects; it does not create input information or establish cheaper acquisition. A software function computing coherence from complex amplitudes is not a physical apparatus measuring those amplitudes.

Open operational/physical question. No physical sensor, field identification, phase-reference instrument or calibrated measurement budget is supplied. SPR-01's adopted population-error assumption is not an error model for vectors, S, A, W or complete coherence. Generic complex-input inversion, complete-snapshot dynamical closure, finite-noise reconstruction, physical implementation and computational advantage remain unestablished. Numerical validity outside the executed fixtures is not globally certified.

SGO-01 exact specialization/deduction. On the declared positive family, exact intensity monotonicity and the inherited signed/coherence inverses identify amplitude, while the entire axial-area channel vanishes. For the published population collision, normalization discards the raw-scale difference and diagonal-only observation misses opposite signed coherence still present in S/I . For the static control v , projection specifically hides a nonzero component of C , recovered by Γ . For u and iu , even the full mathematical snapshot loses common phase, whereas the complete face vectors distinguish the states. These are precise, different mechanisms of invisibility; the observation contract must be named before assigning the loss.

Source identities, evidence access and publication status

The original SPR-01 scientific execution used commit

6794a72e934a5383547c2507d8c4891346cddf7a.

SGO-01 execution used the later SPR-01 publication HEAD

dd6c83cbf5bab228ea4fa4c2b90384d8b70f9772.

This paper's publication commit is a subsequent packaging identity, not a replacement for either execution identity. Relevant sources match the SGO-01 bindings. Native dynamics SHA-256 is

ed1af486a2de5176057d49a795bf32f83705f91507a7f108402623a095d535c4.

The original SGO-01 reference used Python 3.11.15, NumPy 2.4.4, SymPy 1.14.0 and mpmath 1.3.0; the saved-data review used Python 3.13.5 and mpmath 1.3.0. These describe finite numerical records, not assumptions of the exact proofs.

The public package supplies this self-contained derivation, the two unchanged v02 PNGs and a claim/evidence crosswalk. Full raw results, the large CSV, machine-bound verifiers, private review bundles and preservation logs remain undistributed. Reported execution evidence has therefore not been presented as turnkey public replay. The proofs and stated examples are explicit in the paper; the source notes identify the origin and access boundary of the finite evidence.

References

- [1] Tri-Octagon project. *Native Response Geometry and Spectral Circulation in the Tri-Octagon Map*. Repository research publication v0.1, 6 October 2026. The adopted decoder and discrete connection, fibre closure criterion, regular complex-coherence closure and regular real-Gram closure are in the `observables_closure` source. The real-Gram proof supplies the inherited $O(2)$ fibre argument. Commit-pinned source.
- [2] Tri-Octagon project. *Geometry-Attached Axial Observables, Period-Doubling, and Certified Entrance Capture in the Tri-Octagon Map*. Paper G, v0.1.1. Sections 2–3 supply pair/area conventions, attachment reconstruction and accounting identities. No bifurcation or capture certificate is rerun or extended here. Commit-pinned source.
- [3] H. Frímann Halldórsson. *Strength-to-Pattern Conversion and Readout Identifiability in the Tri-Octagon Native Kernel*. v0.1, author-authorized repository research publication, 7 October 2026. Fixed family, intensity, population collision and inverses; Section 9 retains the RFO-02 negative result and direct comparators. Project-internal review, not external peer review. Commit-pinned publication source.
- [4] Tri-Octagon project. *Cycle-Covering Dynamics of a Three-State Nonlinear Kernel*. Paper A, publication v1.0. Section 1 and Section 6.1 give the defining recurrence and zero convention. Source pinned to the original SPR-01 execution revision.
- [5] Tri-Octagon project. Scientific implementation at SGO-01 execution revision dd6c83c: `geometry.py`, `face_state.py`, `axial_observables.py`, `readouts.py`, `z_diagnostics.py`, `dynamics.py` and `api.py`. Cited for existing definitions and execution paths. Commit-pinned source directory.
- [6] Tri-Octagon project. *K0 axial observation extension v0.1*. Existing contract, Domains and guarantees. Its whole-snapshot common-phase wording is qualified in Section 8; the protected contract itself is not rewritten by this publication. Commit-pinned contract.
- [7] Tri-Octagon project. *SGO-01: Native state to existing geometric observation*. Completed project-internal study, 7 October 2026. Report, frozen protocol, exact/numerical results and original verifier; full operational artifacts are not distributed with this paper. Accepted mathematical statements are proved here and finite outcomes are identified as retained evidence.
- [8] GPT. *SGO-01 saved-evidence and visualization review addendum*. Project-internal review, 7 October 2026; not external peer review. Independent 110-digit audit of saved scientific data and visual acceptance of geometry-only v02. Its supplied set lacked the comparison PNG; that image was Codex-inspected and was subsequently included in the owner’s explicit v02 publication authorization. Private review files are not distributed.

AI assistance and responsibility. AI assistance from GPT and Codex was used in mathematical derivation, verification, manuscript preparation, and project-internal review. The recorded reviews and reciprocal checks were conducted within the project; they do not constitute external peer review. Hilmir Frímann Halldórsson approved this repository edition and is responsible for its content and publication.